

Design of a lightweight robotic arm for kiwifruit pollination

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ABSTRACT

Robotic simulation is an important method for verifying robot design parameters and the feasibility of various schemes. We simulated and analysed the effectiveness of the design parameters of a robotic arm for kiwifruit pollination and verified the rationality of the robotic arm's movement during pollination. First, the basic parameters and working space of the robotic arm for kiwifruit pollination were identified based on a trellis system that supports kiwifruit vines. MATLAB software was employed for the simulation to prove that the working space of the lightweight robotic arm for pollination can satisfy the operating requirements. Appropriate joint motors were selected based on the maximum theoretical torque on each joint of the robotic arm obtained through the analysis. SolidWorks software was used to verify the strength of the robotic arm materials and joint connectors, and the operating environment of the robotic arm and MATLAB software were applied to simulate and analyse the feasibility of trajectory planning for pollination. Based on the simulation results, we built a prototype of the robotic arm and a motion control system. The experiment indicated that the maximum operating speed of this lightweight pollination robotic arm was $\omega_{\max} = 0.105$ rad/s, the pollination success rate was 85%, the average pollination time of a single flower was 5 s, and the pollination efficiency was 78 min/mu, which was significantly higher than the 86 min/mu of artificial brush pollination. The lightweight pollination robotic arm satisfied the automatic operation requirements of kiwifruit pollination.

1. Introduction

In 2020, the kiwifruit planting area in China reached 193,000 ha, with an annual output of 2.291 million tons, which was the largest in the world (Zhang et al., 2016). In particular, Shaanxi and Sichuan have vast stretches of kiwifruit trees and produce a high yield of this fruit.

Kiwifruit trees are typically dioecious plants. They do not self-pollinate as they produce functional unisexual flowers owing to the pollen abortion of the female flower and ovary degeneration of the male flower (Meroi Arcerito et al., 2021; Borghezian et al., 2011). For natural propagation, a certain quantity of male kiwifruit trees must be planted in the orchard, and pollination through insects and wind is required. However, climate change and a limited number of insect pollinators could lead to malformed or small fruits owing to inadequate pollination, thereby decreasing fruit quality and yield (Broussard et al., 2021; Estravis-Barcala et al., 2021). Therefore, in the process of breeding kiwifruit trees, artificial pollination methods, such as cross-pollination, brush pollination, and manual or electric pollen sprayers, are often used to compensate for the low kiwifruit yield and quality resulting from

natural pollination (Saez et al., 2019; Yang and Miyako, 2020); however, such processes require high labour intensity (Abbate et al., 2021; Chechetka et al., 2017; Mu et al., 2018; Oh et al., 2021).

In view of the shortcomings of artificial pollination, such as high labour intensity, many experts and scholars have proposed the use of automated pollination operations to fully, uniformly, and efficiently improve the pollination rate and fruit bearing rate of plants (Castro et al., 2021; Singh et al., 2021; Yao et al., 2019) and to make up for the shortcomings of artificial pollination operations (Castro et al., 2021; Singh et al., 2021; Yao et al., 2019). Examples include the use of unmanned aerial vehicles (UAVs) to pollinate wheat, corn, and other field crops by means of wind (Del Cerro et al., 2021; Li et al., 2015). The kiwifruit pollination robot developed by Henry Williams et al. can locate kiwifruit flowers in trellis cultivation mode while driving at a speed of 3.5 km/hr, and the success rate of single-flower pollination is 79.5% (Williams et al., 2020).

In recent years, the kiwifruit planting area in China has continued to expand and the price of pollen has continued to rise. The price of kiwifruit pollen per gram exceeds 200 yuan. At the same time, the cost of

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agricultural labour in China is rising (Gonzalez-de-Santos et al., 2020; Williams et al., 2019). High pollen prices, labour prices, and labour shortages have placed high requirements on the accuracy, effectiveness, and pollen consumption of artificial pollination (Sakamoto et al., 2017; Sakamoto et al., 2009). Therefore, China needs a large amount of intelligent kiwifruit flower pollination equipment (Williams et al., 2020).

At present, most of the research on the automatic pollination operation of kiwifruit has focused on the perception of target information of kiwifruit flowers, focusing on the automated recognition of kiwifruit flowers (He et al., 2019; Sa et al., 2016) and the detection of pollination positions (Ji et al., 2020; Mohd et al., 2020; de Luna et al., 2019). However, very few reports focus on the use of physical robotic arms for the automatic pollination of kiwifruit. Existing research reports on the operation of robotic arms not only reflect the low operating efficiency and accuracy of the robotic arms (Sepulveda et al., 2020; Williams et al., 2019), but also show that the robotic arms used are large and have a complex structure, high weight, high inertia, and high manufacturing cost (Sepulveda et al., 2020; Williams et al., 2019).

The research object of this study was kiwifruit grown in a natural environment. Our aim was to address the problems in previous research focusing on robotic arm operations. To this end, we designed a lightweight kiwifruit pollination robotic arm. The rationality of its configuration design parameters meets the size requirements of the kiwifruit pollination space in the trellis cultivation mode, and the manufacturing material nylon 6 can meet the rigidity and strength requirements of the lightweight robotic arm, thereby improving its durability. In the process of the continuous pollination of the kiwifruit flowers under the trellis, the speed and acceleration of each joint of the robotic arm did not change abruptly when the robotic arm moved at the highest angular velocity $\omega_{\max} = 0.105 \text{ rad/s}$, which satisfied the requirements of the kiwifruit pollination robot with high motion accuracy and smooth operation. Compared with traditional pollination programs such as artificially assisted pollination, the contradiction analysis shows that the robotic arm automatic pollination program proposed in this study has a pollination success rate of more than 85%, pollination takes 78 min/mu, and its pollination efficiency is significantly higher than that of the artificially assisted pollination of 86 min/mu, which proves the feasibility of the robotic arm automated pollination program.

2. Principle of automated pollination by a robotic pollinator

Automated kiwifruit pollination was conducted at the kiwifruit test station at Northwest A&F University, Mei County, Shaanxi Province, China ($4^{\circ}7'39''\text{N}$, $107^{\circ}59'50''\text{E}$, 648 m above sea level). As is shown in Fig. 1, kiwifruit trees cultivated on a trellis were arranged in rows. These trees, with a line spacing of 4.0 m and a column spacing of 3.0 m, were supported by a T-shaped trellis (1.8 m in height) built with steel pipes



Fig. 1. A kiwifruit orchard with a trellis system.

and iron wires. Regular kiwifruit flowers were distributed in clusters along iron wires on top of the trellis. These flowers, which were generally 1,650–1,800 mm above the ground, differed by an average of 150 mm in height. Thus, the robotic pollinator had sufficient space to conduct mobile operations. Prior to pollination, the flowers were thinned. The kiwifruit vines after flower thinning are shown in Fig. 2. Each vine usually had 4–6 flowers on it. The flowers were sparsely distributed, with little overlap, and grew in various positions and orientations.

A schematic diagram of the operation of the automatic kiwifruit pollinator is shown in Fig. 3. The pollinator consisted of a wheeled chassis, robotic arm, and binocular camera system. The binocular camera can take pictures of the trellis top in real time and identify kiwifruit flowers and uses the principle of parallax imaging (Jiang et al., 2020; Jiang et al., 2019; Li et al., 2014) to obtain the three-dimensional coordinates of the flowers in the vision system. After obtaining the three-dimensional coordinates of the flowers in the vision system, they were converted to the three-dimensional coordinates of the robotic arm operating system prior to sending them to the main controller to drive the robotic arm for pollination.

According to the growth and distribution characteristics of kiwifruit flowers cultivated on trellis, and the characteristics of the robotic pollinator, and in consideration of the flexibility and position continuity of pollination operations, the mobile operation mode of the robotic kiwifruit pollinator used the intermittent walking trajectory shown in Fig. 4. The pollinator moved intermittently under a 4-m-wide trellis. The cubic area for a single pollination was 0.5 m in length, 0.5 m in width, and 150 mm in height. The pollinator started pollinating from origin O on the lower left of the aisle, as is shown in Fig. 4. After the flowers above the current stop were pollinated, the pollinator continued to move forward following the trajectory shown in Fig. 4. The pollinator stopped moving every 50 cm to repeat the pollination. This process was repeated.

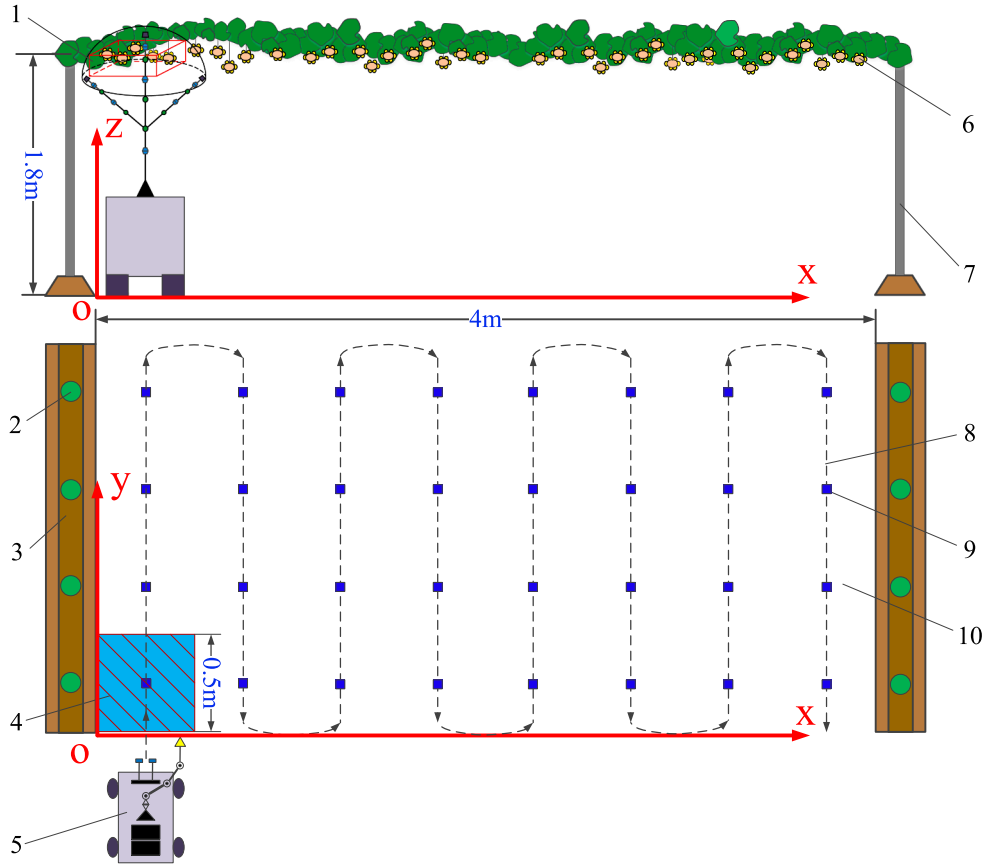
3. Simulation analysis of the working space of the robotic arm for the pollination of kiwifruit

3.1. Theoretical analysis

During the pollination operation of the robotic arm, the end spraying device of the robotic arm must be able to reach any location in the operation area, and can perform pollination operations in sequence according to the position and orientation of the flowers. As the end of the robotic arm with six degrees of freedom can reach any point in its working space (Wang and Hou, 2018), we selected an articulated robotic arm with six degrees of freedom as the pollinator arm, as shown in Fig. 5. In particular, the waist rotation joint on the S-axis, the pitching joint of the upper arm on the L-axis, and the pitching joint of the forearm on the U-axis controlled the pollinator at the end so that the activity space at the end of the forearm can cover the pollination area of the kiwifruit flower. The rotating joint on the R-axis, the pitching joint on the B-axis, and the rotating joint on the T-axis can control the orientation of the end



Fig. 2. Distribution of kiwifruit flowers.



1. Umbrella-shaped working space for pollination; 2. kiwifruit tree; 3. zones of kiwifruit trees distributed in columns;

4. location for a single pollination; 5. robotic pollinator; 6. kiwifruit flowers; 7. kiwifruit trellis; 8. trajectory of the robotic pollinator;

9. stop of the robotic pollinator; 10. pollinating space (ridge)

Fig. 4. Schematic diagram of the intermittent mobile operation of the robotic pollinator. 1. Umbrella-shaped working space for pollination; 2. kiwifruit tree; 3. zones of kiwifruit trees distributed in columns; 4. location for a single pollination; 5. robotic pollinator; 6. kiwifruit flowers; 7. kiwifruit trellis; 8. trajectory of the robotic pollinator; 9. stop of the robotic pollinator; 10. pollinating space (ridge).

$$\begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ s_6 & c_6 & 0 & a_2 s_2 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

If $c_1 = \sin\theta_1$, $c_{23} = \cos(\theta_2 + \theta_3)$, $s_1 = \sin\theta_1$, and $s_{23} = \sin(\theta_2 + \theta_3)$, then the end matrix is as follows:

Based on the parameters of the robotic arm and kinematic parameters, the Monte Carlo method (Yue et al., 2018; Zhu et al., 2017) was used to analyse the working space of the robotic arm. Several rotation angles of different joints were selected to calculate the coordinates of the

$${}^0T = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 {}^5T_6 = \begin{bmatrix} -c_6[s_5c_1s_{23} + c_5(s_1s_4 - c_4c_1c_{23})] & -s_6[s_5c_1s_{23} + c_5(s_1s_4 - c_4c_1c_{23})] & c_5c_1s_{23} - s_5(s_1s_4 - c_4c_1c_{23}) & d_6[c_5c_1s_{23} - s_5(s_1s_4 - c_4c_1c_{23})] \\ -s_6(s_1c_4 + s_4c_1c_{23}) & -c_6(s_1c_4 + s_4c_1c_{23}) & c_5c_1s_{23} - s_5(s_1s_4 - c_4c_1c_{23}) & +a_2c_2c_1 + d_4c_1s_{23} \\ -c_6[s_5c_1s_{23} - c_5(c_1s_4 + c_4s_1c_{23})] & s_6[s_5c_1s_{23} - c_5(c_1s_4 + c_4s_1c_{23})] & c_5s_1s_{23} + s_5(c_1s_4 + c_4s_1c_{23}) & d_6[c_5s_1s_{23} + s_5(c_1s_4 + c_4s_1c_{23})] \\ +s_6(c_1c_4 - s_4s_1c_{23}) & +c_6(c_1c_4 - s_4s_1c_{23}) & c_5s_1s_{23} + s_5(c_1s_4 + c_4s_1c_{23}) & +a_2c_2s_1 + d_4s_1s_{23} \\ s_4s_6s_{23} - c_6(s_5c_{23} + c_4c_5s_{23}) & s_6(s_5c_{23} + c_4c_5s_{23}) & c_5c_{23} + c_4s_5s_{23} & d_4c_{23} - a_2s_2 + d_6[(c_5c_{23} - c_4s_5s_{23})] \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

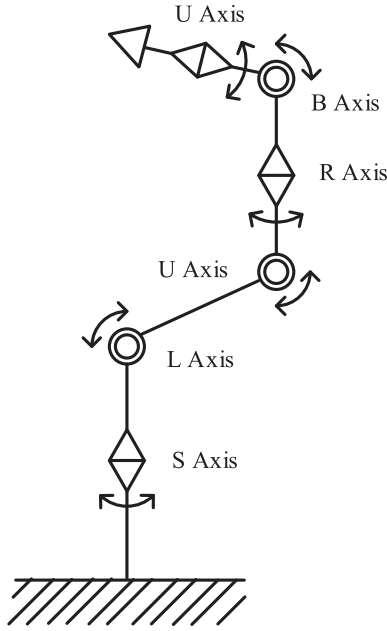


Fig. 5. Schematic diagram for the mechanism of the robotic arm used for pollination.

end using forward kinematic equations, and all the points in the working space were collected into a set. The end point of the robotic arm relative to the base coordinate system 0 was expressed as follows:

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = T_6^0 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} d_6[c_5s_1s_{23} - s_5(s_1s_4 - c_4c_1c_{23})] + a_2c_2c_1 + d_4c_1s_{23} \\ d_6[c_5s_1s_{23} + s_5(c_1s_4 + c_4s_1c_{23})] + a_2c_2s_1 + d_4s_1s_{23} \\ d_4c_{23} - a_2s_2 + d_6[(c_5c_{23} - c_4s_5s_{23})] \\ 1 \end{bmatrix} \quad (1)$$

MATLAB was employed to simulate all positions that the end could reach. A total of 10,000 random values within the angle range of each rotational joint were substituted into Eq. (1). Through the forward kinematics solution, 10,000 working points used to describe the working space of the robotic arm were obtained. As is shown in Fig. 8, when $-37.5 \text{ cm} < x < 37.5 \text{ cm}$, $-37.5 \text{ cm} < y < 37.5 \text{ cm}$, and $40 \text{ cm} < z < 60 \text{ cm}$, the working space of the robotic arm can include all the points in the pollination zone. According to the Monte Carlo method (Yue et al., 2018; Zhu et al., 2017) analysis, the designed operation space size of $750 \text{ mm} \times 750 \text{ mm} \times 200 \text{ mm}$ met the requirements of pollination work; that is, the end of the robotic arm could reach any location in the operation space area for pollination, which proves that the operation space and the

length of the robotic arm lever were appropriately designed. As the actual height of kiwifruit flowers was approximately 1,800 mm, a chassis of 1,200 mm in height was required to ensure that the robotic arm could reach the kiwifruit flowers.

4. Simulation analysis of the forces on the robotic arm

4.1. Theoretical analysis

When the arm levers above the waist of the robotic arm reached the position shown in Fig. 9, the joints had the largest moment of force and moment of inertia at the centre of rotation. A coordinate system was established, with the base of the robotic arm being the origin, the direction of the upper arm being the X-axis, the direction of the shoulder joint being the Y-axis, and the vertical upward direction being the Z-axis. The centres of mass of both the upper arm and forearm levers were in the centre of the arm lever, and that of each joint was in the geometric centre of each joint, all indicated by black dots. In particular, if the mass of the upper arm lever $m_0 \approx 250 \text{ g}$, that of the elbow's pitching joint m_1

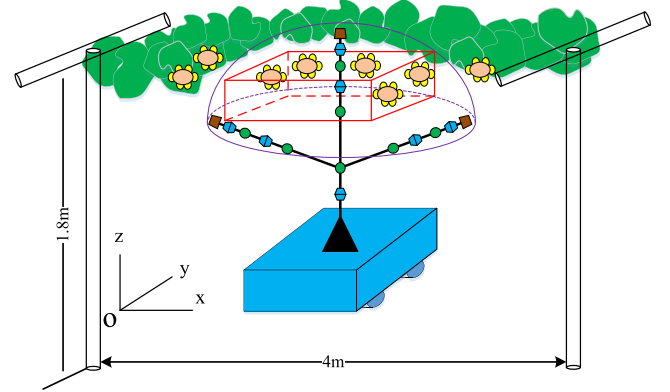


Fig. 7. Schematic diagram of the working space of the robotic arm.

Table 1

D-H parameters of the robotic arm.

Joint i	$\alpha_i (^\circ)$	$a_i (\text{m})$	$d_i (\text{m})$	$\theta_i (^\circ)$
1	-90°	0	0	θ_1
2	0	0.42	0	θ_2
3	90°	0	0	θ_3
4	-90°	0	0.40	θ_4
5	90°	0	0	θ_5
6	0	0	0.05	θ_6

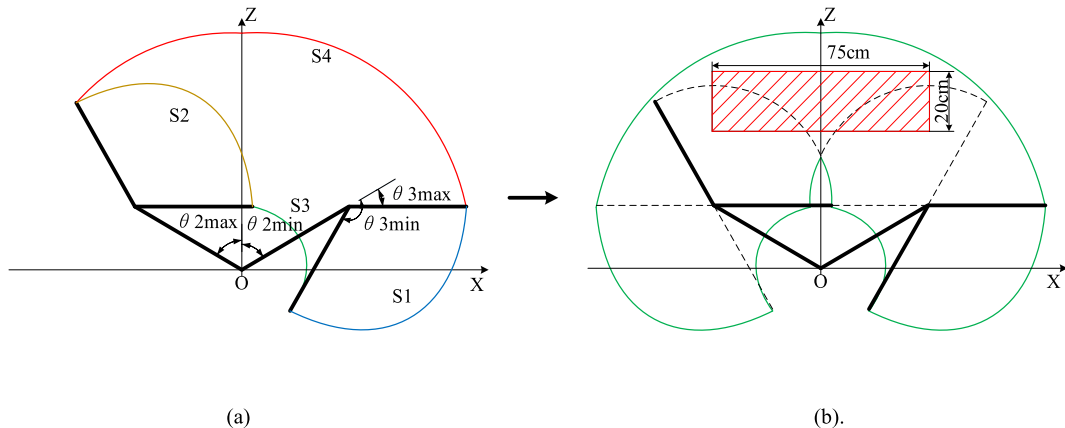


Fig. 6. Working space of the robotic arm.

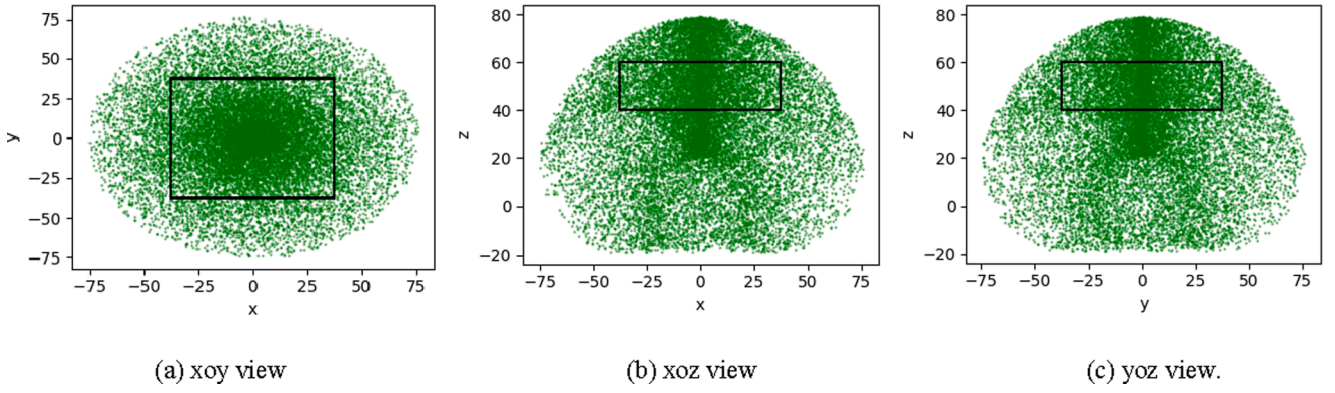


Fig. 8. Simulation results of the working space.

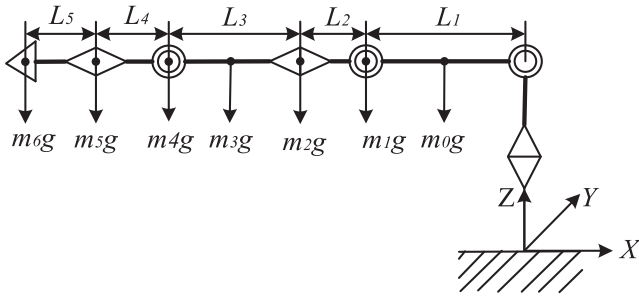


Fig. 9. Force analysis of the robotic arm.

≈ 855 g, that of the elbow's rotation joint $m_2 \approx 735$ g, that of the forearm lever $m_3 \approx 200$ g, that of the wrist's pitching joint $m_4 \approx 339$ g, that of the wrist's rotational joint $m_5 \approx 230$ g, that of the end executor $m_6 \approx 100$ g, the lengths of the joint connecting rods $L_1 = 420$ mm, $L_2 = 30$ mm, $L_3 = 370$ mm, $L_4 = 20$ mm, and $L_5 = 50$ mm, and the acceleration of gravity $g = 9.8$ m/s², then the moments of force on the shoulder joint, elbow joint, and wrist pitching joint are as follows:

$$\begin{aligned}
 T_1 &= m_0 g \frac{L_1}{2} + m_1 g L_1 + m_2 g (L_2 + L_1) + m_3 g \left(\frac{L_3}{2} + L_2 + L_1 \right) + m_4 g (L_3 + L_2 + L_1) + m_5 g (L_4 + L_3 + L_2 + L_1) \\
 &\quad + m_6 g (L_5 + L_4 + L_3 + L_2 + L_1) = 13.999594 \text{ N}\cdot\text{m} \\
 T_2 &= m_2 g L_2 + m_3 g \left(\frac{L_3}{2} + L_2 \right) + m_4 g (L_3 + L_2) + m_5 g (L_4 + L_3 + L_2) + m_6 g (L_5 + L_4 + L_3 + L_2) = 3.36385 \text{ N}\cdot\text{m} \\
 T_3 &= m_5 g L_4 + m_6 g (L_5 + L_4) = 0.10388 \text{ N}\cdot\text{m}
 \end{aligned} \tag{2}$$

Table 2
Parameters for the motors of each joint.

Component	Rated Voltage	Rated Current	Static Moment	Reduction Ratio	Mass
Waist Motor	12 V	1.3 A	0.27 N·m	1:5	400 g
Shoulder Motor	12 V	1.3 A	0.27 N·m	1:15	600 g
Elbow Motor	12 V	1.3 A	0.27 N·m	1:5	400 g
Yaw Motor at Elbow	12 V	1.2 A	0.6 N·m	/	350 g
Pitching Motor at Wrist	12 V	0.6 A	0.18 N·m	/	200 g
Rolling Motor at Wrist	12 V	0.6 A	0.12 N·m	/	150 g

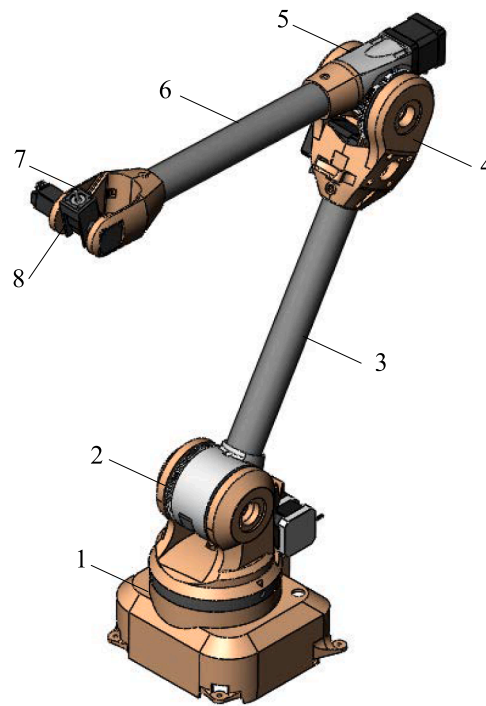
When the robotic arm is operating, the pollinator must be accurately controlled; however, the requirement for the movement speed of the robotic arm end is not strict. Therefore, planetary gears were used as speed reducers for each joint. They were driven using a 1:5 synchronous belt. We used 42 planetary reduction stepper motors (42HS3413A4CE-XG) in the waist, shoulder, and elbow joints of the robotic arms, and applied a 42 stepper motor (42HB48-401A) at the yaw joint at the wrist as well as 28 stepper motors (28HB-401A) at the pitching and rolling joints at the wrist. The parameters are listed in Table 2.

4.2. Simulation verification of the joint stress

The three-dimensional modelling software SolidWorks was used to design the lightweight robotic arm for kiwifruit pollination (Fig. 10), which included four parts: the base, upper arm, forearm, and wrist joints. Both the upper arm and forearm arm levers were made of carbon fibre tubes. Except for the standard parts, such as the motor, reducer, and screws, the key parts of the robotic arm were 3D printed with nylon 6. The characteristic parameters of the nylon 6 and carbon fibre tubes

are listed in Tables 3 and 4. To ensure that the nylon 6 material met the strength requirements of the lightweight robotic arm for the production material, it was necessary to carry out stress-strain analysis and strength checks on the key components of the robotic arm made of nylon 6 to ensure that the stress and strain of each key component were far less than the strength limit of the nylon 6 material during the operation of the robotic arm, thereby guaranteeing the durability of nylon 6.

When the robotic arm is operating, the shoulder and elbow joints bear a large torque. The shoulder joint torque is borne by the shoulder supports and rotating parts, whereas the elbow joint torque is sustained by the elbow supports and rotating parts. The finite element simulation plug-in of SolidWorks Simulation was applied to conduct stress-strain analysis and strength checks for key components of the robotic arm, and torques of 20 N·m and 4 N·m were applied to the shoulder rotating parts and shoulder supports, respectively. The stress-strain simulation results are shown in Fig. 11. At this moment, the shoulder rotating parts had a



1. Waist joint; 2. shoulder joint; 3. upper arm; 4. elbow joint; 5. yaw joint at the wrist; 6. forearm; 7. pitching joint at the wrist; 8. rolling joint at the wrist

Fig. 10. 3D Model of the lightweight robotic arm used for the pollination of kiwifruit. 1. Waist joint; 2. shoulder joint; 3. upper arm; 4. elbow joint; 5. yaw joint at the wrist; 6. forearm; 7. pitching joint at the wrist; 8. rolling joint at the wrist.

Table 3
Characteristic parameters of nylon 6.

Mechanical Property	Glass Transition Temperature	Density	Melting Point	Elasticity Modulus	Tensile Strength
Value	223 °C	1.13 g/cm ³	220 °C	3,400 MPa	70–84 MPa
Print Parameters	Print Thickness	Filling Density	Printing Temperature	Filling Rate	Average Wall Thickness
Value	0.15 mm	100%	260 °C	20–40 mm/s	2.5 mm

Table 4
Performance parameters of carbon fibre tubes.

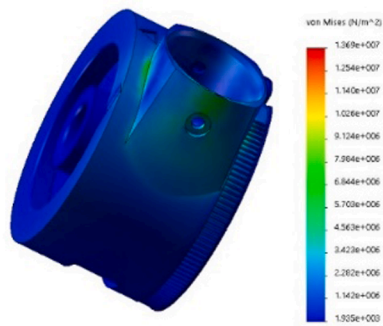
Performance	Gram Weight (g/m ²)	Tensile Strength	Elasticity Modulus	Elongation	Rated Current	Density
Parameters	300	3,400 MPa	240 GPa	1.7%	4.7 A	1.5–2.0 g/cm ³

maximum stress of 13.69 MPa and a maximum strain of 0.001035, with a maximum stress of 5.425 MPa and a maximum strain of approximately 0.0003801. This means that the stress and strain of the rotating shoulder parts and supports were both less than the ultimate strength of nylon 6. Similarly, under a torque of 10 N·m, the elbow rotating parts had a maximum stress of 13.02 MPa and a maximum strain of 0.0009942; under a torque of 2 N·m, the elbow supports had a maximum stress of 2.649 MPa and a maximum strain of approximately 0.0002426. This means that the stress and strain of the elbow rotating parts and supports were both less than the ultimate strength of nylon 6. It shows that the production materials of the key components of the pollination robotic arm were properly selected, and the strength met the operation requirements.

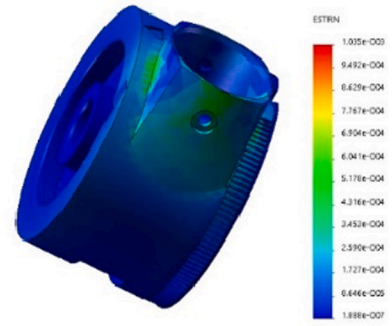
5. Simulation analysis on the trajectory of the robotic arm for pollination

5.1. Theoretical analysis

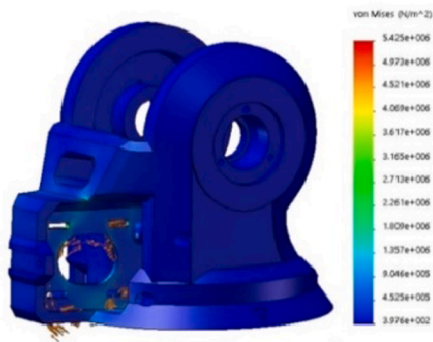
In order to reduce the impact force on the pollination robotic arm during the operation and improve the motion stability, it is required that the speed and acceleration curves of each joint of the robotic arm must be continuous and smooth, and there should be no sudden changes in speed and acceleration. In the scenario where the robotic arm consecutively pollinates two flowers, the sequential pollination path for the end of the robotic arm is designed according to the trellis system supporting kiwifruit vines, as shown in Fig. 12. By taking pictures with the binocular camera in the information perception system and locating the kiwifruit flowers on the top of the trellis, we can obtain the three-dimensional coordinates of all kiwifruit flowers in the working space and determine the pollination sequence according to the spatial



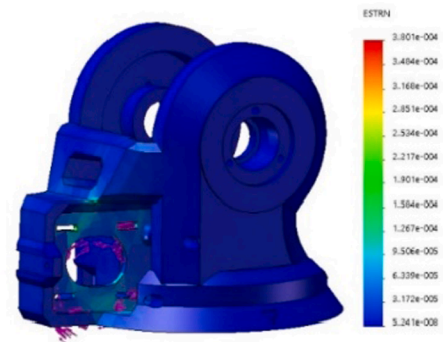
(a) Stress diagram of the shoulder rotating parts



(a) Strain diagram of the shoulder rotating parts



(a) Stress diagram of the shoulder supports



(a) Strain diagram of the shoulder supports

Fig. 11. Stress-strain diagram for the parts of the shoulder joints.

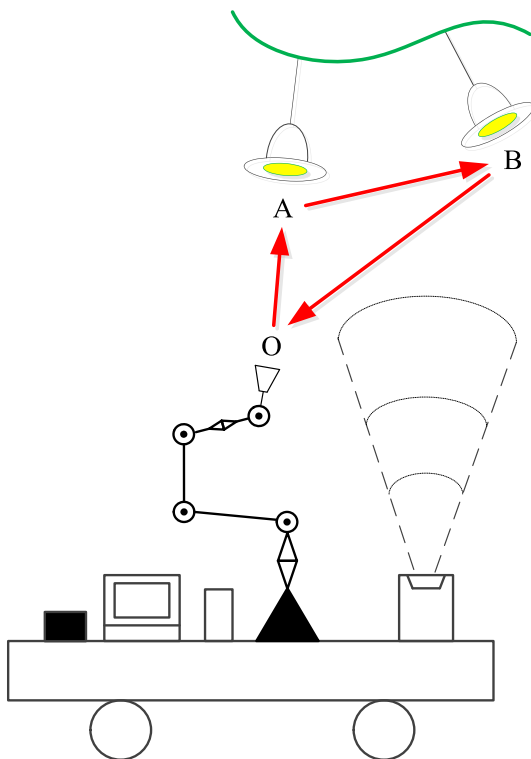


Fig. 12. Schematic diagram of the sequential pollination trajectory of the end of the robotic arm.

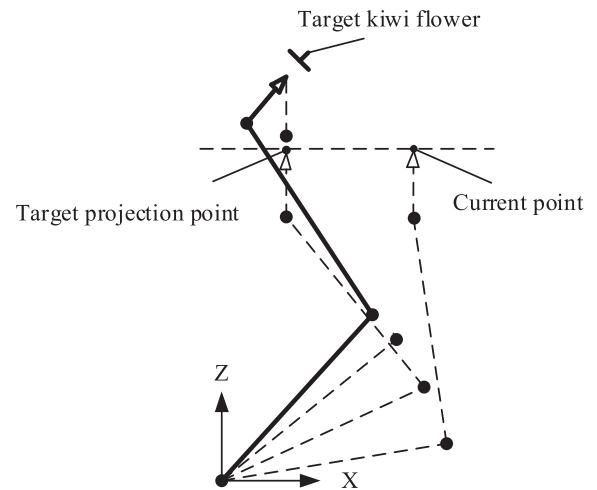


Fig. 13. Schematic diagram of the actions of pollination.

positions of the flowers. The end of the robotic arm moved from starting point O to the first pollinating point in the working space for pollination. Upon completion, it sequentially pollinated other flowers until all kiwifruit flowers in a single working space were pollinated. Finally, the end of the robotic arm returned to the starting point O. In this study, the process of continuous pollination using a robotic arm can be divided into three stages.

- 1) The end of the robotic arm moves from starting point O to target flower A.
- 2) The end of the robotic arm moves from target flower A to target

flower B.

3) The end of the robotic arm moves from target flower B to starting point O.

At each stage of robotic pollination, quintic polynomial interpolation (Duditza et al., 1987) was used to design the trajectory at the end of the robotic arm. To allow the pollinator at the end of the robotic arm to rapidly and accurately reach the kiwifruit flower, the trajectory of the end of the robotic arm for pollination at each stage was designed with four actions, as shown in Fig. 13: i.e. horizontal linear movement, vertical linear movement, orientation change, and return. The literature (Kondo et al., 1996a; Kondo et al., 1996b) shows that in tomato picking, it takes 5 s for a robotic arm to move from the initial position to the target tomato, with a maximum joint angular velocity of 0.105 rad/s. Based on the literature, the maximum joint angular velocity of the robotic arm for pollination was designed as $\omega_{\max} = 0.105$ rad/s. The maximum range of motion of the joints was 180° . Subsequently, for the simulation analysis of the trajectory of the robotic arm for pollination, when the joints of the robotic arm move at the maximum angular velocity $\omega_{\max}$, the speed and acceleration of the robotic arm do not experience a sudden change.

When the end of the robotic arm moves from the starting point to the pollination point in the working space, the speed at the end of the robotic arm and the angular velocity of each joint can be expressed using the Jacobian matrix of the robotic arm:

$$\begin{bmatrix} v \\ \tilde{\omega} \end{bmatrix} = J(q)\dot{q} \quad (3)$$

The robotic arm uses rotating joints that can be expressed with the following formulas:

$$\begin{bmatrix} v \\ \tilde{\omega} \end{bmatrix} = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} \dot{q}_i \quad (4)$$

$$J_i = \begin{bmatrix} z_i \times {}^i p_n^0 \\ z_i \end{bmatrix} \quad (5)$$

where J_i is the i th column of the Jacobian matrix, ${}^i p_n^0$ represents the position vector of the origin in the coordinate system of the robotic arm end relative to the coordinate system $\{i\}$ in the base coordinate system, and z_i denotes the unit vector of the z -axis of the coordinate system $\{i\}$ in the base coordinate system. The robotic arm model used in this study satisfies the following formulas:

$$z_i = {}^0 T_i \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad (6)$$

$${}^i p_n^0 = {}^{i-1} T_i \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} \quad (7)$$

where p_x , p_y , and p_z represent the positions of the ends of the robotic arm in the end-coordinate system. The Jacobian matrix of the robotic arm was obtained according to the rotation matrix of each coordinate system. The Jacobian matrix of the robotic arm is related to the joint angle and end position. The joint angle can be obtained based on the end position when the end posture is determined. At this moment, the Jacobian matrix is associated only with the end position.

$$J = J(p_x, p_y, p_z) \quad (8)$$

When the robotic arm moves along the designed linear path, the linear parameter equation for its movement from starting point $o = (x_o, y_o, z_o)$ to pollinating point $e = (x_e, y_e, z_e)$ in the working space is:

$$\begin{cases} x = x_o + (x_e - x_o)t \\ y = y_o + (y_e - y_o)t \\ z = z_o + (z_e - z_o)t \end{cases} \quad (0 \leq t \leq 1) \quad (9)$$

Its velocity equation is a function of the robotic arm end with respect to time t :

$$\begin{cases} x = x_o + \frac{s(v(t), t)(x_e - x_o)}{\sqrt{(x_e - x_o)^2 + (y_e - y_o)^2 + (z_e - z_o)^2}} \\ y = y_o + \frac{s(v(t), t)(y_e - y_o)}{\sqrt{(x_e - x_o)^2 + (y_e - y_o)^2 + (z_e - z_o)^2}} \\ z = z_o + \frac{s(v(t), t)(z_e - z_o)}{\sqrt{(x_e - x_o)^2 + (y_e - y_o)^2 + (z_e - z_o)^2}} \end{cases} \quad (10)$$

According to the end of the robotic arm, we can obtain the function of the Jacobian matrix with respect to time to obtain the function of each joint velocity with respect to time.

$$\dot{q}(t) = J^{-1}(x(t), y(t), z(t)) \begin{bmatrix} v(t) \\ \tilde{\omega} \end{bmatrix} \quad (11)$$

5.2. Simulation verification of the working trajectory

Quintic polynomial interpolation was used to design the spatial trajectory of the robotic arm joint. We set the joint angle at a certain moment as $\theta(t)$, the start and end times as t_0 and t_f , respectively, and the start and end joint angles as θ_0 and θ_f , respectively. The constraints are as follows:

$$\begin{cases} \theta(t_0) = \theta_0 \\ \theta(t_f) = \theta_f \\ \dot{\theta}(t_0) = 0 \\ \dot{\theta}(t_f) = 0 \\ \ddot{\theta}(t_0) = 0 \\ \ddot{\theta}(t_f) = 0 \end{cases} \quad (12)$$

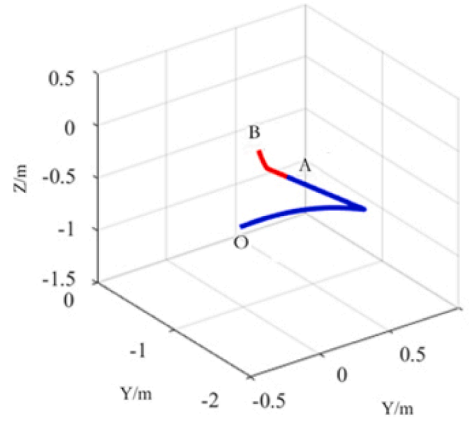
According to constraint Equation (12), we obtain the following equation set:

$$\begin{cases} \theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \\ \dot{\theta}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4 \\ \ddot{\theta}(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 \end{cases} \quad (13)$$

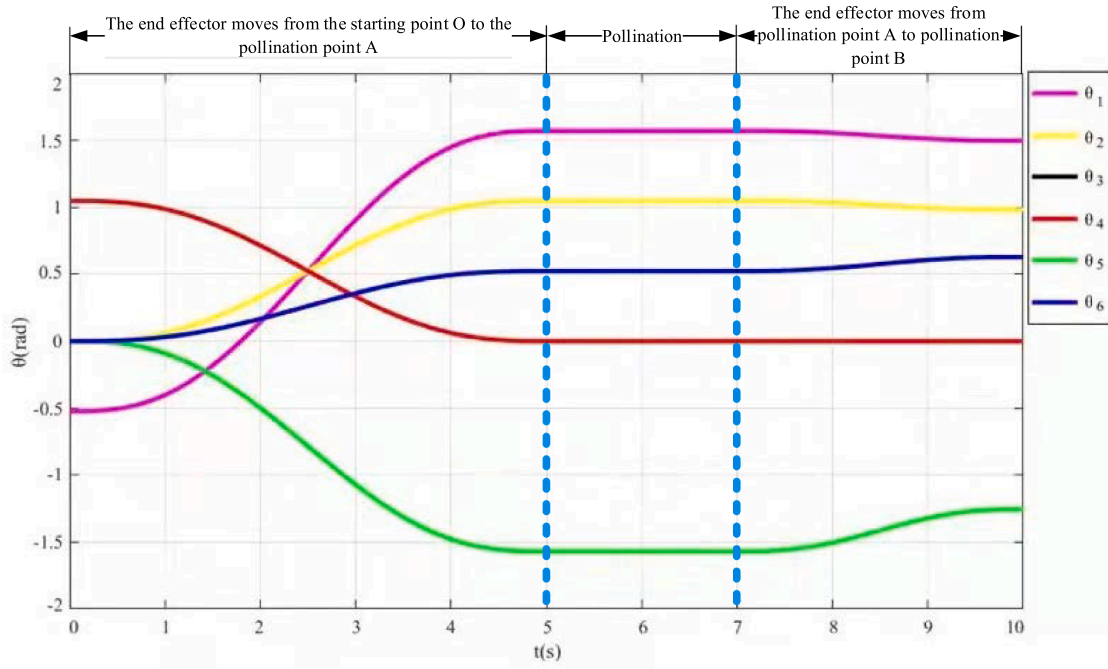
Two adjacent points in the trajectory are considered as the start and end positions. The start and end joint angular velocities were v_0 and v_f , respectively, and the start and end joint angular accelerations were a_0 and a_f , respectively.

$$\begin{cases} \theta(t_0) = \theta_0 = a_0 \\ \theta(t_f) = \theta_f = a_0 + a_1 t_f + a_2 t_f^2 + a_3 t_f^3 + a_4 t_f^4 + a_5 t_f^5 \\ \dot{\theta}(t_0) = v_0 = a_1 \\ \dot{\theta}(t_f) = v_f = a_1 + 2a_2 t_f + 3a_3 t_f^2 + 4a_4 t_f^3 + 5a_5 t_f^4 \\ \ddot{\theta}(t_0) = a_f = 2a_2 \\ \ddot{\theta}(t_f) = a_f = 2a_2 + 6a_3 t_f + 12a_4 t_f^2 + 20a_5 t_f^3 \end{cases} \quad (14)$$

We substitute Equation (14) into Equation (13) to obtain the trajectory planning equation based on quintic polynomial interpolation:



(a) Motion trajectory at the end



(b) Position curve of each joint.

Fig. 14. Simulation result of joint space trajectory planning.

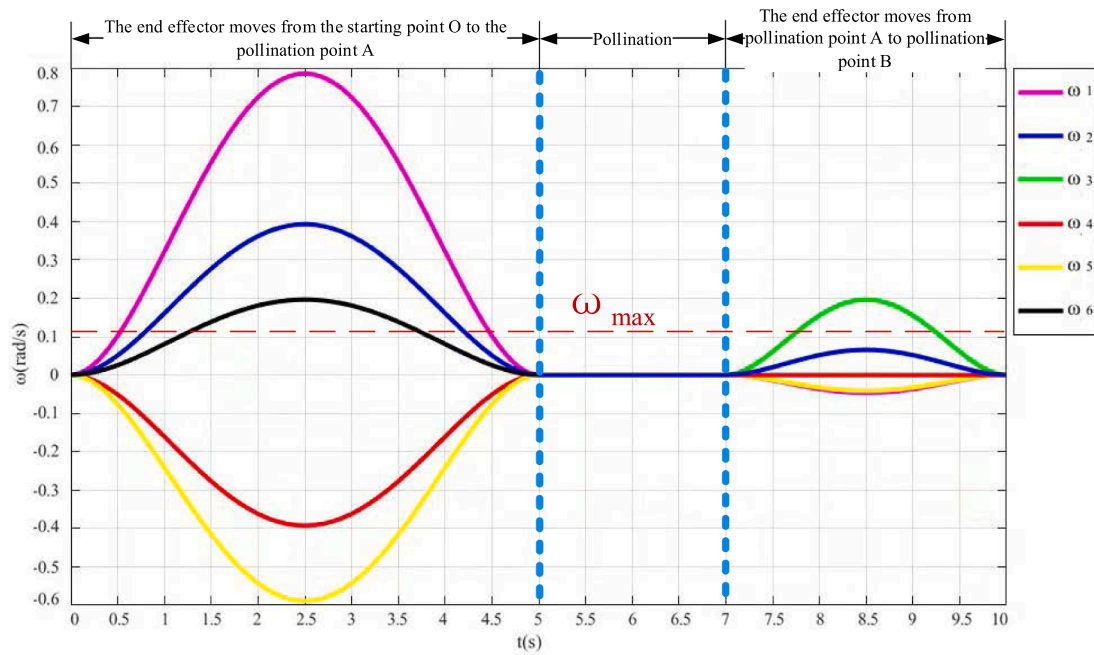
$$\begin{cases}
 \theta_o = a_0 \\
 \dot{\theta}(t) = a_1 \\
 \ddot{\theta}(t) = 2a_2 \\
 20\theta_f - 20\theta_o - \left(8\dot{\theta}_f + 12\dot{\theta}_o\right)t_f + \left(\ddot{\theta}_f - 3\ddot{\theta}_o\right)t_f^2 = 2a_3t_f^3 \\
 30\theta_o - 30\theta_f + \left(14\dot{\theta}_f + 16\dot{\theta}_o\right)t_f - \left(2\ddot{\theta}_f - 3\ddot{\theta}_o\right)t_f^2 = 2a_4t_f^4 \\
 -12\theta_o + 12\theta_f - \left(6\dot{\theta}_f + 6\dot{\theta}_o\right)t_f + \left(\ddot{\theta}_f - \ddot{\theta}_o\right)t_f^2 = 2a_5t_f^5
 \end{cases} \quad (15)$$

According to the analysis mentioned above, joint space trajectory planning with quintic polynomial interpolation was conducted based on the robot toolbox in MATLAB. As shown in Fig. 14(a), the end of the robotic arm moved from starting point O to the vicinity of pollinating point A in the working space. The end executor performed pollination during which the joints of the robotic arm remained still. At the end of

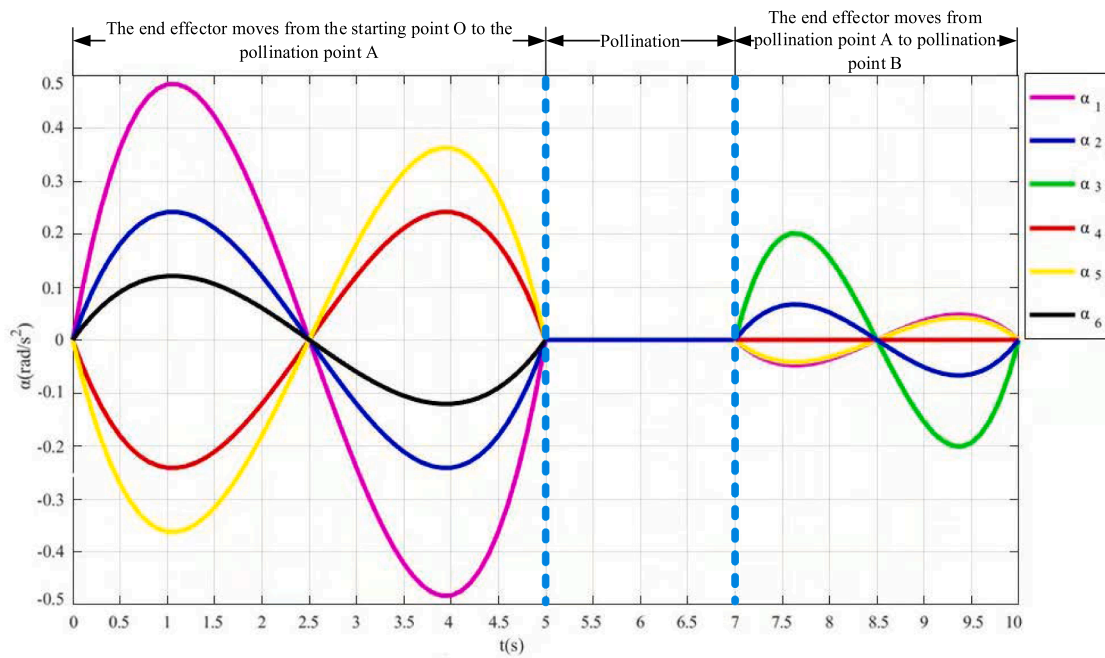
pollination, the robotic arm moved to pollinating point B. The position, velocity, and acceleration curves of each joint of the robotic arm during this process are shown in Fig. 14(b), (c), and (d), respectively.

Based on the simulation results, under joint space trajectory planning, even if the joint movement speed $\omega > \omega_{\max}$, the acceleration does not change suddenly.

The three-dimensional model of the robotic arm designed using SolidWorks was imported into MATLAB for trajectory planning simulation in the Cartesian coordinate system. The Simscape module of the Simulink plug-in was used to build a rigid body tree. In the Cartesian coordinate system, the trajectory of the robotic arm was planned to lead the end of the robotic arm to move from the starting point to the end point along a straight line, as shown in Fig. 15(a). The changes in the end position and orientation during the movement are shown in Fig. 15(b) and 15(c), respectively. According to the simulation results, the end position and orientation curves in the trajectory planning simulation under the Cartesian coordinate system were smooth and stable without sudden changes.



(c) Velocity curve of each joint



(d) Acceleration curve of each joint

Fig. 14. (continued).

6. Experimental verification

6.1. Establishment of the experimentation platform

A prototype of the robotic arm was built according to the simulation experimental data mentioned above, as shown in Fig. 16. The body of the robotic arm was made of nylon 6 via 3D printing. The end executor used an atomiser (YSVCO-CC-6510-ss, Dongguan Shaou Spray System Co. LTD, Dongguan, Guangdong, China) and a binocular camera (KS1A552-D, Kingcent Technology Co. LTD, Shenzhen, Guangdong, China), which were installed on the base.

The control system of the pollination robotic arm also included an

industrial personal computer (IPC) (D12120P551, Shenzhen Hongda-feng Electronics Co. LTD, Shenzhen, Guangdong, China), motion controller (Battleship V3, Xingyi Electronic Technology Co. LTD, Guangzhou, Guangdong, China), six motor drivers (Emm42A S-tepper_V3.x, Chengdu Aikong Electronic Technology Co. LTD, Chengdu, Sichuan, China), and six stepper motors (42HS3413A4CE-XG, Dongguan Xunkong Automation Technology Co. LTD, Dongguan, Guangdong, China). In particular, the motor driver can provide feedback on the position and rotation speed of the motor to the motion controller in real time. The total manufacturing cost of the robotic arm prototype was RMB 7932, and the total power was 75.6 W.

The workflow of the control system is shown in Fig. 17. A binocular

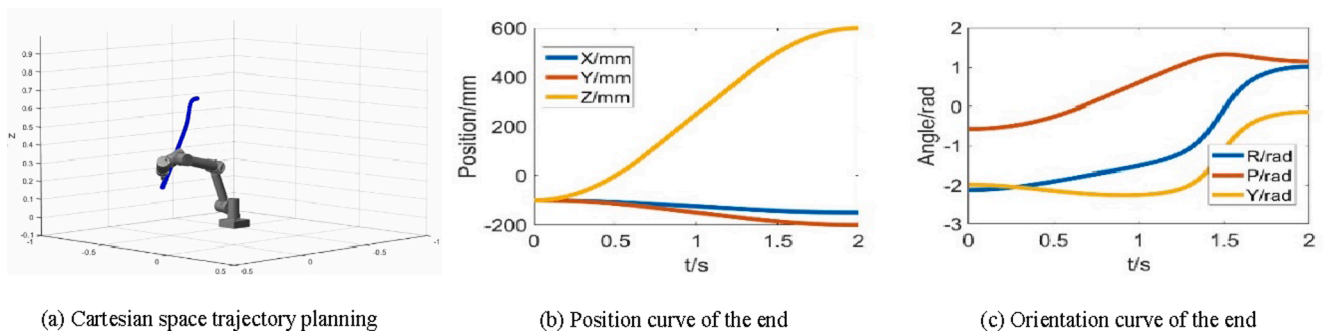
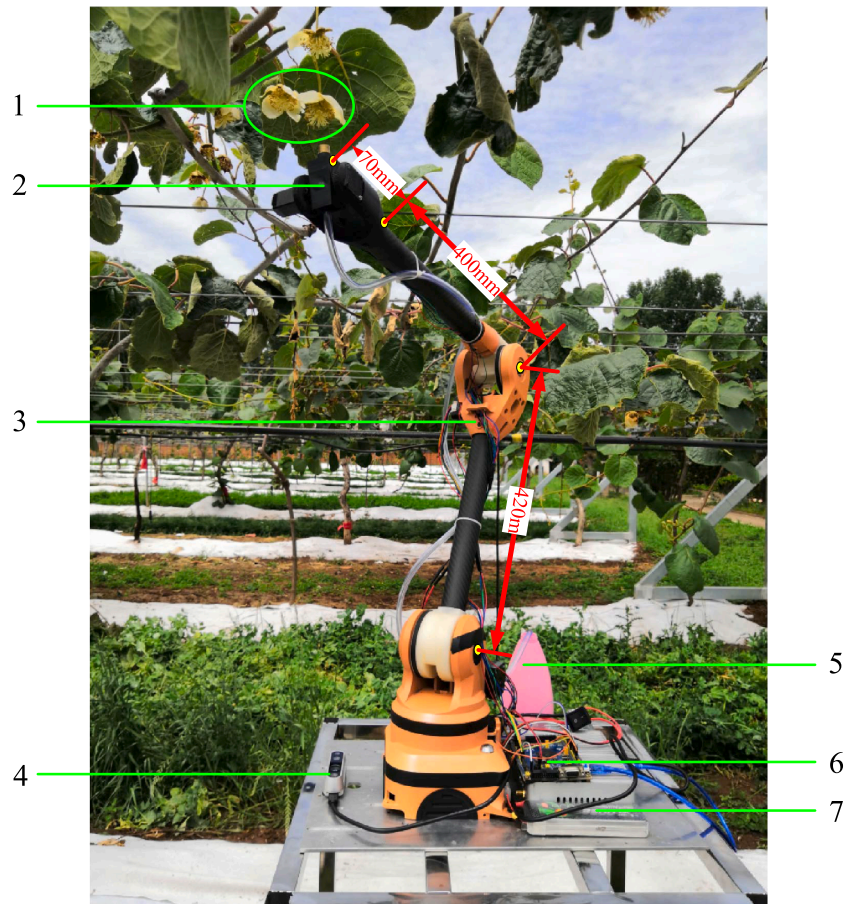


Fig. 15. Simulation results of Cartesian space trajectory planning.



1. Kiwifruit flowers to be pollinated; 2. end executor (sprayer); 3. lightweight robotic arm; 4. binocular camera; 5. pollen pot; 6. master controller; 7. IPC

Fig. 16. The prototype of the robotic arm used for pollination. 1. Kiwifruit flowers to be pollinated; 2. end executor (sprayer); 3. lightweight robotic arm; 4. binocular camera; 5. pollen pot; 6. master controller; 7. IPC.

camera was used to acquire scene images of kiwifruit flowers in the field in real time. The distributed operating environment ROS embedded within the IPC was utilised to process images, obtain the orientation information of target flowers, and conduct trajectory planning (Meng et al., 2016). It could compute the interpolation points of the robotic arm in real time and then send the interpolation point information to the master controller, which read the current position information of each motor and process the information. The control information was sent to the motor driver of each joint to control the joint rotation and change the position and orientation of the robotic arm end. The encoder provided

feedback on whether each joint motor reached the designated position to ensure that the sprayer at the end of the robotic arm conducted point-to-point pollination of the target flowers.

Among them, the key point of accurately judging the flower position and orientation is to obtain the centre of gravity of the pistil and the centre of gravity of the flower contour. Through the two centre points, the central axis representing the growth direction of the flower could be obtained, that is, the flower position and orientation. First, we acquired kiwifruit flower images with a binocular camera and determined the pistil contour of the flower in the images of the left and right eyes using a

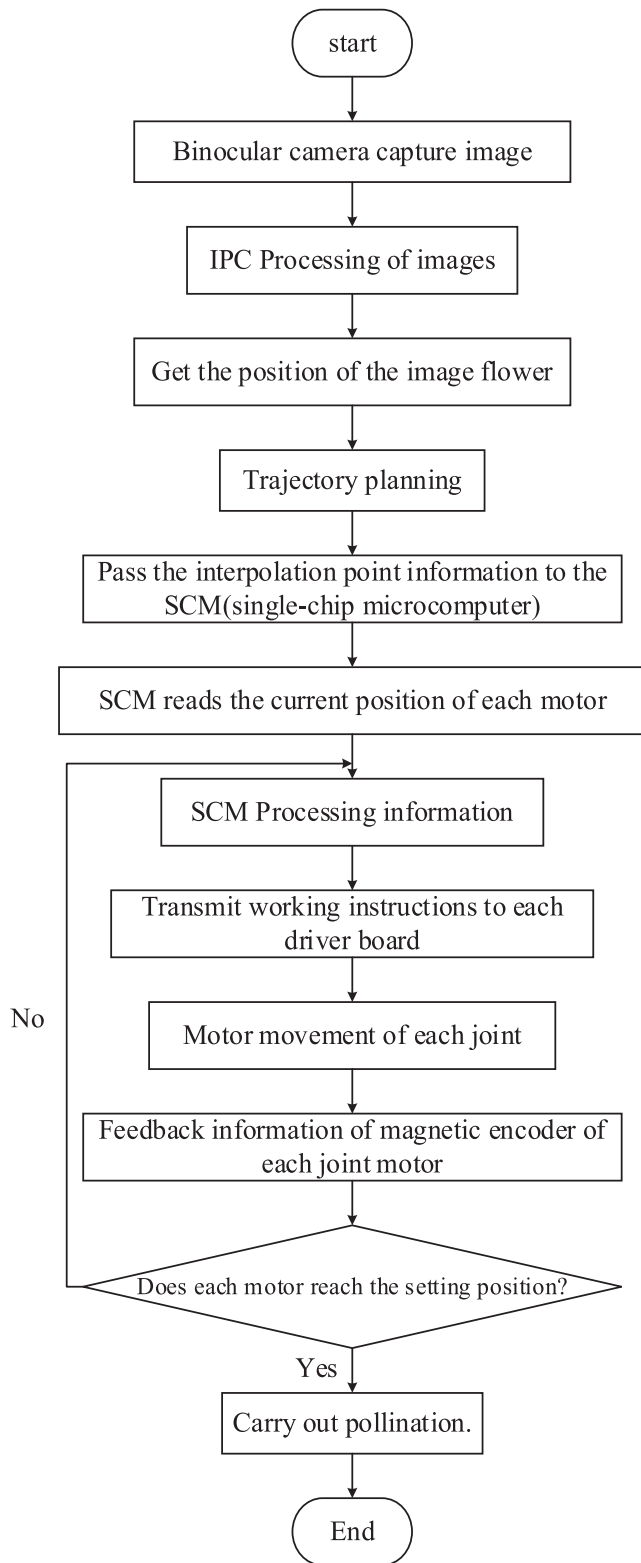


Fig. 17. Control flow of the robotic arm.

contour detection algorithm. Next, we applied a binarisation method in fitting to obtain the centre of gravity of the pistil contour in the images of the left and right eyes and their 2D coordinates. Then, based on the distance between the optical centres of the two cameras and the 2D coordinates of the pistil's centre of gravity in images for the left and right eyes, as well as the binocular parallax imaging principle (Jiang et al., 2020; Jiang et al., 2019; Li et al., 2014), we determined the spatial 3D

coordinates of the pistil's centre of gravity using the least-squares method (Jing Song and Wen Wang, 2014). Using the same method, we obtained the spatial 3D coordinates of the centre of gravity for the minimum circumscribed circle of the petal. Since pollination was carried out on female kiwifruit flowers that bloomed for 1–3 days, and the lifespan of kiwifruit female flowers is 2–6 days, the female kiwifruit flowers will not be greatly deformed due to withering during the pollination process. Therefore, the outline of the female kiwifruit flower will not be greatly distorted, and its centre of gravity will not change with age.

To reduce the computation of image processing, after using the YOLOv4 detection model to detect kiwifruit flowers, we drew a coordinate zone region of interest (ROI) for the predictive frame of target flowers as boundaries of the detecting contour and contour zone's centre of gravity. We obtained the contour boundary of the minimum circumscribed circle of the pistil through detection, and the centre of gravity of the pistil contour by fitting according to the contour boundary. Similarly, we detected and generated the petal's minimum circumscribed circle contour and obtained the petal contour's centre of gravity using a contour detection algorithm provided by OpenCV.

The process of detecting the centre of gravity of the pistil and petal contour is shown in Fig. 18. Firstly, the kiwifruit flower image is pre-processed, and after the kiwifruit flower image is grayscaled, a binary image representing the characteristics of the kiwifruit petal area and the pistil area is obtained using local adaptive threshold segmentation (Goh et al., 2018). Next, we used a boundary-tracking algorithm to detect contours in the binary image (Ren et al., 2002). We screened the contours based on the geometric property of the target contour, obtained the pistils' and petals' minimum circumscribed circle contour information, and removed irrelevant contours (Zhang et al., 2020). Then, according to the minimum circumscribed circle contour information of both the pistil and petal, we obtained the centres of gravity of the two contours by fitting, and calculated their spatial coordinates according to the binocular parallax principle.

The results for the pistil's centre of gravity and flower contour detection and the legend of the fitting flower axis are shown in Fig. 19 (a)–(e), as shown in the figure, the orange rectangle represents the selected contents of petal outline frame, the blue rectangle represents the selected contents of pistil outline frame, the yellow rectangle

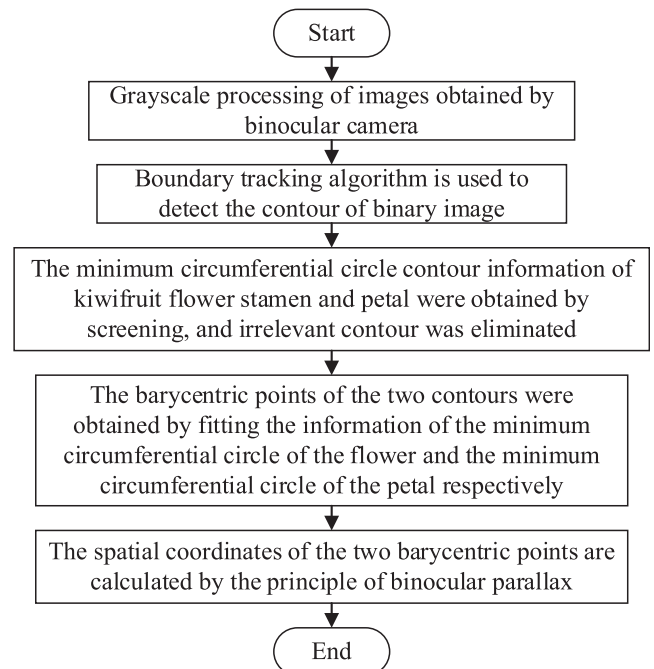


Fig. 18. Process of identifying centre of gravity of pistil and flower contour.

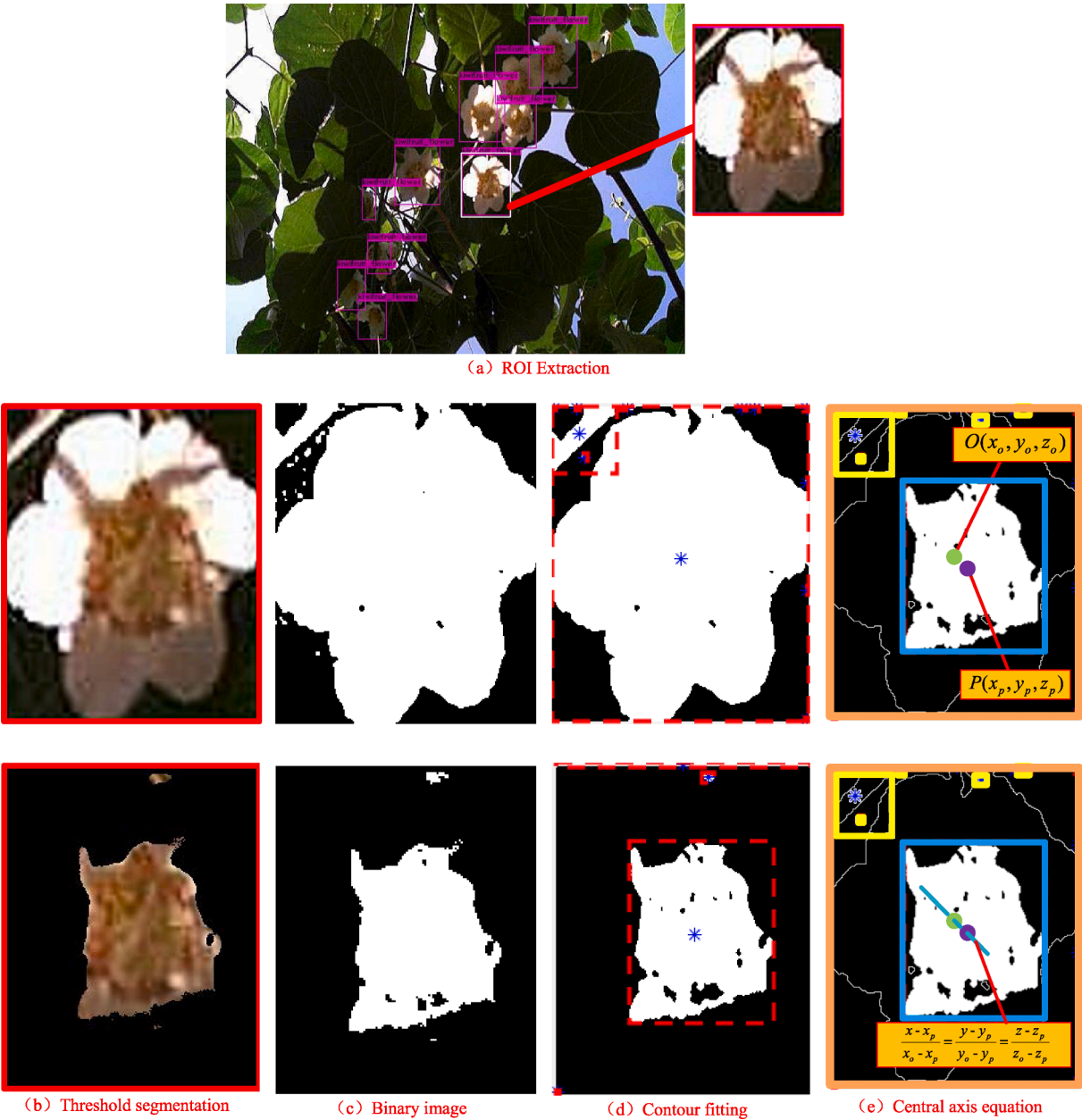


Fig. 19. Instance diagram for central axis of kiwifruit flower position and orientation.

represents the irrelevant contents of interference recognition, the green dot is the petal centre of gravity, the purple dot is the pistil centre of gravity, and the blue central axis represents the flower posture central axis.

6.2. Experimental results

A lightweight pollination robotic arm was used to pollinate kiwifruit flowers on a trellis system to verify the performance of the robotic arm at the kiwifruit test station in Northwest A&F University, Mei County, Shaanxi Province, China (4°7'39"N, 107°59'50"E, 648 m above sea level).

Table 5
Statistics of the robotic pollinator’s targeted pollination performance test.

Pollination Zone	Performance Index				Total Number of Flowers
	Success Rate of Pollination/%	Success Rate of Flower Detection/%	Total Time of Pollination/s	Average Pollination Time of a Single Flower/s	
Zone 1	86	92	105	5	21
Zone 2	87	93	115	5	23
Zone 3	82	91	100	5	20
Mean	85	92	107	5	

We randomly selected pollinating positions, drew pollination test zones, and divided each pollinating zone into $2\text{ m} \times 2\text{ m}$ squares. Before pollination, we first summarised the number of kiwifruit flowers in each pollination test zone and then started using the pollinator, which employed a binocular camera to identify the orientations of kiwifruit flowers for ‘point-to-point’ pollination. The results of the pollination performance tests are listed in Table 5. The success rates of the robotic arm were 86%, 87%, and 82% in the selected zones 1–3, respectively, with an average value of 85%; the pollination duration of a single flower was 5 s for each zone, averaging 5 s; the total pollination duration was 105 s, 115 s, and 100 s, respectively, averaging 107 s. The ‘success rate of flower detection’ is defined as the proportion of the number of successfully detected flowers to the total number of flowers in the entire area. The ‘success rate of pollination’ is defined as the proportion of the number of flowers successfully pollinated to the total number of flowers in the pollination area when the robotic arm pollinates the successfully detected flowers. However, the success rate of pollination does not reflect whether the spray nozzle has deviated from the centre of gravity because of an inappropriate nozzle position while pollinating the detected flowers, in which case pollination is considered to have failed.

The ‘total duration of pollination’ is defined as the total time from the beginning of flower detection by the information perception system to the completion of continuous pollination by the controlled robotic arm of all the detected flowers in the pollination zone, according to their positions and orientations. The ‘average pollination duration’ of a single flower is defined as the total time from when the robotic arm moves to the vicinity of a detected flower, according to the given operation instruction to the adjustment of the nozzle position to pollinate the target flower to move to the next flower upon completion of pollination.

During the specific pollination process, the robotic arm was started from the initial position, and moved to the position near the kiwifruit flower to be pollinated according to the planned optimal path. When the optimal pollination position of the kiwifruit flower was reached, the kiwifruit flower was targeted for pollination. After pollination was completed, the robotic arm moved to the best pollination position of the next flower and performed targeted pollination until all flowers were pollinated, and the robotic arm returned to the initial position. Among them, the local operation process of the robotic arm to continuously pollinate kiwifruit flowers is shown in Fig. 20 (a) – (f) respectively. Take the example of consecutively pollinating 2 clusters of kiwifruit flowers. First, the robotic arm moved from the initial position to the position near the kiwifruit flower in the area ① according to the optimal path, and when it reached the optimal pollination position of a flower in the area ①, the target pollination was performed on the kiwifruit flower. Then, when all the kiwifruit flowers in area ① had been pollinated, moved to area ② and repeated the previous work process. Finally, the robotic arm was reset and returned to the initial position.

During the pollination process of the robotic arm, the Hall sensors installed on each joint can detect the angular displacement of each joint and joint activity time. According to the recorded data of joint angular displacement and joint activity time, we calculated the displacement, velocity, and acceleration curves of each joint of the robotic arm and drew the corresponding curves. The speed and acceleration curves of each joint during the continuous pollination of flowers in pollination zone 1 are shown in Fig. 21(a) and 21(b), respectively. Even if the angular velocity of a certain joint of the robotic arm changed slightly during the moving operation (for example, the velocity curve in the interval from 27 s to 31 s shown in Fig. 21(a) had a slight mutation), the acceleration curves of the corresponding joints in this time period can still keep it smooth and not abrupt. The curve between 27 s and 31 s is shown in Fig. 21(b). This proves the stability of the robotic arm during the actual operation.

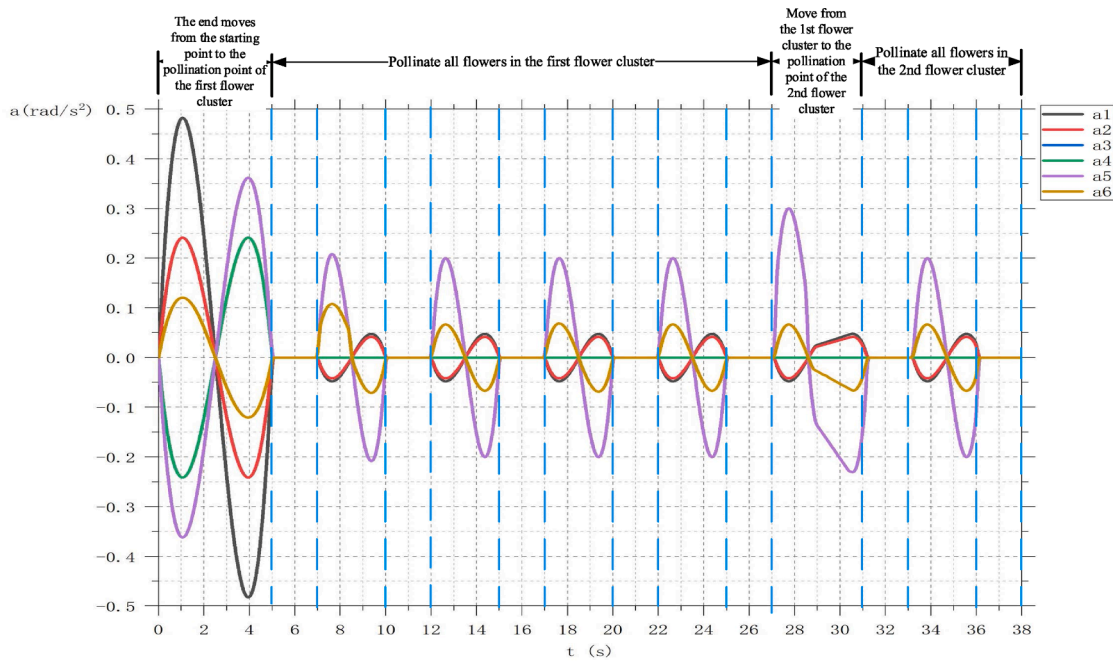
7. Discussion

A comparison of Figs. 21 and 14 shows that the actual pollination process of the robotic arm is consistent with the simulation result of joint space trajectory planning shown in Fig. 14; that is, the velocity and acceleration curves of each joint are generally smooth and stable. No mutations were observed. However, as shown in Fig. 21(a), while the velocity curve in the interval from 27 s to 31 s has a slight mutation, the acceleration curve of the robotic arm in the corresponding time interval in Fig. 21(b) also shows signs of a slight mutation. This may be because when the robotic arm completes the pollination task of a cluster of flowers and moves to the next flower cluster, the angular velocity of the joints at this time is relatively unstable, and the robotic arm may encounter obstacles, such as branches and vines, which are affected by occlusion. Owing to the influence of occlusion during the operation of the robotic arm, it was forced to sharply adjust the pose and change the working path for obstacle avoidance, resulting in a slight sudden change in the angular acceleration of each of its joints. However, in general, the movement of the robotic arm during the actual operation was stable. When operating at a maximum speed of 0.105 rad/s , the acceleration curve of each joint remained smooth and did not change abruptly, while it was not affected significantly by obstacles in the operating environment.

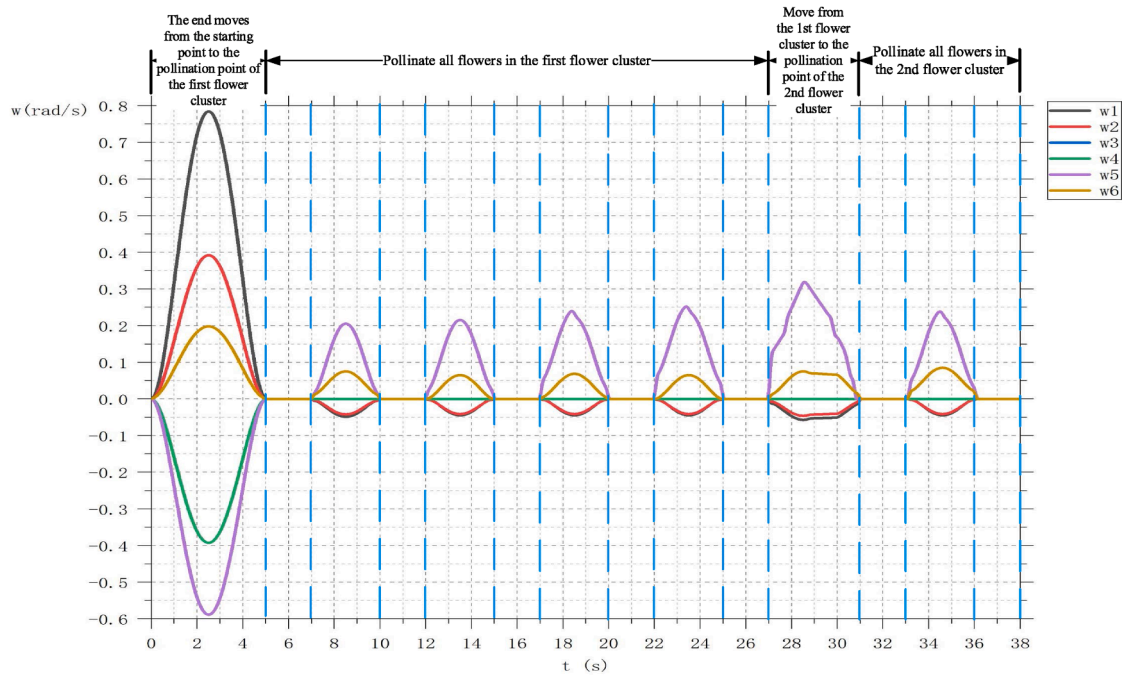


Figure 20 Local pollination process of the robotic arm

Fig. 20. Local pollination process of the robotic arm.



(b) Acceleration curve of each joint



(a) Velocity curve of each joint

Fig. 21. Changes of motion parameters of each joint during the actual pollination process of the robotic arm.

In order to verify the feasibility of the robotic arm automated pollination program, we compared the robotic arm automated pollination with conventional pollination programs, such as manual-assisted pollination (Shu et al., 2015) and electric pollinator pollination (Shu et al., 2015). The results are presented in Table 6. The definition of pollination success rate is the same as that in Section 6.2. As a comparative reference for the robotic arm automated pollination program, the success rates of both manual pollination and electric pollinator pollination were 100%. Pollination efficiency refers to the time taken to pollinate an acre of a

kiwifruit tree.

According to the information shown in Table 6, the pollination success rate of the lightweight robotic arm was lower than that of manual brush point pollination and electric pollinator pollination. The pollination success rate was only 85% because during the pollination process, the accuracy rate of the binocular camera system used to guide the robotic arm to kiwifruit flowers may have been reduced owing to strong light irradiation, which affects the pollination of the target by the robotic arm, resulting in pollination failure. This can lead to a decrease

Table 6

Comparison of different pollination programs.

Pollination Method	Equipment (Product Name)	Pollination Success Rate	Pollination Efficiency
Artificial Pollination	Writing Brush (Shu et al., 2015)	100%	86 min/mu
Electric Pollination (Semi-automatic Pollination)	BS-CFSF-1和BS-CFSF-2 (Shu et al., 2015)	100%	64 min/mu
Automated Pollination	Lightweight Robotic Arm	85%	78 min/mu

in the pollination success rate. The pollination efficiency of the lightweight robotic arm was lower than that of the electric pollinator and higher than that of the artificial brush pollination because artificial brush pollination, as a traditional manual pollination method that does not rely on mechanised equipment, leads to a decrease in the energy of operators, a slowdown in pollination speed, and a decrease in pollination efficiency as pollination work continues to advance. During the entire automated pollination process, the robotic arm can maintain a stable operation efficiency of 78 min/mu, which is higher than the 86 min/mu operation efficiency of manual brush point pollination. However, when the robotic arm operates on the optimal path, it needs to perform information calculation processing, such as obstacle avoidance and collision prevention, which will prolong the operation time. The electric pollinator omits the calculation and information processing, thereby reducing the pollination duration. Omitting too much information processing and calculation time makes the pollination efficiency of the lightweight robotic arm lower than that of the 64 min/mu electric pollinator.

However, the success rate of robotic arm automated pollination was slightly lower than that of manual brush point pollination and electric pollinator pollination. However, owing to its stable operation efficiency, robotic arm automatic pollination could effectively relieve the work pressure of operators during the pollination process of manual brush point pollination and electric pollinators and provide the greatest convenience for pollination operations. At the same time, artificial brush point pollination can easily damage the stamen stigma, resulting in a decrease in the fruit setting rate and influencing the pollination effect (Shu et al., 2015). Electric pollinators mostly require large-scale spraying, which can easily cause pollen waste (Shu et al., 2015). Therefore, based on a comprehensive analysis, automatic pollination using robotic arms remains the optimal solution.

8. Conclusion

In this study, we simulated and designed a lightweight robotic arm for kiwifruit pollination. First, the configuration, basic parameters, and working space of the robotic arm for pollination were identified according to the operating environment. MATLAB was used to simulate the working space. The results indicated that the design parameters of the robotic arm were appropriate. Subsequently, the forces on the robotic arm were analysed to explore the maximum theoretical moment of the force of each joint. Based on the results, appropriate joint motors were selected, and the SolidWorks simulation was used to analyse the stress and strain of the joint connectors and verify the rationality of the structural design of the joints. Next, according to the pollination environment and the working space, the trajectory control had to meet certain requirements: neither speed nor acceleration could undergo sudden changes when the maximum joint speed planning was adopted. MATLAB was used to simulate trajectory planning and choose the joint space trajectory planning based on quintic polynomial interpolation. Finally, a prototype of the robotic arm was built to verify its practical performance. The results showed that designing a lightweight robotic arm through simulation was feasible and the resulting pollinating performance was excellent. It took 5 s on average to pollinate each kiwifruit

flower, with an average success rate of 85%. When the robotic arm moved at a maximum speed of $\omega_{\max} = 0.105$ rad/s, the end nozzle could reach any position within the working space range of 750 mm × 750 mm × 200 mm, and the speed and acceleration of each joint were guaranteed not to cause mutations. The pollination efficiency reached 78 min/mu, which was significantly higher than that of artificial brush pollination at 86 min/mu. The experiment indicates that with excellent kinematic performance, the robotic arm can meet the requirements of high movement precision and stable operation to achieve point-to-point targeted pollination. The robotic arm for kiwifruit pollination is characterised by a small volume, light weight, low cost, and highly targeted application. It can also be applied to other fields of automatic agricultural production.

The lightweight kiwifruit pollination robotic arm designed in this study has the following significant advantages compared with the robotic arms reported in previous studies:

Compared with the robotic arms used in previous research reports (Shu et al., 2015), the lightweight robotic arm designed in this study is small in size, lightweight, has low inertia, is easier to start and brake quickly, and has lower manufacturing cost; the latter is only RMB 7932, which is far lower than the general price of industrial robotic arms sold in the market, thereby making the lightweight robotic arm more suitable for the agricultural industry with low income per unit area. In addition, the motor and reducer are built into the cavity of the robotic arm, which simplifies the configuration, satisfies the structure of the robotic arm, and improves the protection of the motor circuit.

CRediT authorship contribution statement

Kai Li: Data curation, Investigation, Validation, Writing – original draft. **Yujia Huo:** Data curation, Investigation. **Yinan Liu:** Data curation, Validation. **Yinggang Shi:** Conceptualization, Data curation, Methodology, Supervision, Writing – review & editing. **Zhi He:** Data curation, Investigation, Validation. **Yongjie Cui:** Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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